

Bangladesh University of Textiles
Department of Textile Engineering Management

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Screening of Ring-Spinning Process Parameters
Affecting Yarn Hairiness Index Using a Plackett–Burman Design (Minitab)

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1. Objective

The primary objective of this experiment is to screen six candidate ring-spinning process parameters and identify, with the minimum possible number of trials, which of them have a statistically and practically significant effect on yarn Hairiness Index, using a Plackett–Burman screening Design of Experiments (DoE) in Minitab. Specifically, this study aims to:

1. Evaluate the individual (main) effect of Spindle Speed, Spinning Tension, Traveler Weight, Ring Diameter, Yarn Count, and Twist Multiplier on yarn hairiness using a 12-run Plackett–Burman design.
2. Rank the six factors by their standardized effect on hairiness using a Pareto chart and formal hypothesis tests (ANOVA / t-tests).
3. Separate the ‘vital few’ factors that materially affect hairiness from the ‘trivial many’ that can be fixed at any convenient level.
4. Recommend a focused, higher-resolution follow-up experiment (optimization stage) on only the factors found to be significant, so that experimental effort is not wasted on unimportant variables.

2. Theory / Background

Hairiness is one of the most important yarn-quality attributes for ring-spun yarn: protruding fibres reduce fabric appearance, increase pilling and end-breaks in downstream processes, and can indicate excessive fibre damage or poor spinning-zone control. Hairiness is influenced by a large number of machine settings, so before committing to a full, expensive optimization study it is standard industrial and statistical practice to first run a screening experiment that quickly identifies which of the many candidate parameters actually matter.

2.1 Screening Designs

A screening design is a Design of Experiments strategy used early in an investigation, when many factors (here, six) are suspected of influencing a response but resources do not permit a full $2^6 = 64$ -run (or larger) factorial study. Screening designs sacrifice some information — mainly the ability to cleanly separate two-factor interactions from main effects — in exchange for a very large reduction in the number of runs required, following the Pareto principle that, typically, only a few of many candidate factors turn out to be practically important.

2.2 Plackett–Burman Design

The Plackett–Burman (PB) design is a two-level, resolution III fractional factorial screening design, constructed so that runs are always a multiple of 4. For 6 factors, Minitab uses the standard 12-run Plackett–Burman array (the next size up from the minimum of 8), which gives $(12 - 6 - 1) = 5$ residual degrees of freedom for estimating experimental error even though every factor is studied at only 2 levels. Its key properties are:

- Orthogonality: every pair of factor columns is uncorrelated, so all six main effects can be estimated independently of one another (VIF = 1.00 for every term, as confirmed in the regression output).
- Resolution III: main effects are aliased (confounded) with two-factor and higher-order interactions. A significant main effect could therefore, in principle, actually be driven by an aliased interaction — this is the price paid for using only 12 runs instead of 64.
- Efficiency: only 12 experimental runs are needed to screen 6 factors, versus 64 runs for a full 2^6 factorial — a reduction of over 80% in experimental effort.

2.3 Pareto Chart of Standardized Effects

Once the linear (main-effects) model is fitted, Minitab's Pareto Chart of Standardized Effects ranks the six factors by the absolute value of their standardized effect (t-statistic). A reference line, calculated from Lenth's pseudo-standard-error method at the chosen significance level ($\alpha = 0.05$ in this study), is superimposed: bars that cross this line are candidates for statistically significant effects, while bars that fall short are not distinguishable from noise at the stated confidence level.

3. Process Parameters and Levels

Six ring-spinning process parameters, believed from prior experience to potentially influence yarn hairiness, were selected for screening:

Symbol	Factor	Low Level	High Level	Unit
A	Spindle Speed	12000	16000	rpm
B	Spinning Tension	10	22	cN
C	Traveler Weight	2.5	4.0	mg
D	Ring Diameter	36	42	mm
E	Yarn Count	20	40	Ne
F	Twist Multiplier	3.5	4.5	–

Note: While setting up the Create Screening Design > Factors dialog in Minitab, the Spindle Speed low level was initially mistyped as 120000 rpm (Figure 5). This was identified and corrected to 12000 rpm before the design was generated (Figure 6), giving the valid 12000–16000 rpm range shown above.

3.1 Response (Output) Parameter

Yarn Hairiness Index (H): a smaller-the-better response, since fewer protruding fibres indicate a smoother, higher-quality yarn. Three replicate hairiness readings (H1, H2, H3) were recorded for each of the 12 trials. The H1 replicate values were supplied by the course instructor as the reference dataset for this experiment; the H2 and H3 replicate values were additionally generated by the student to allow the variability/replication aspect of the analysis to be practiced.

4. Process Flow

The overall procedure followed to design, execute, and analyse the screening experiment in Minitab is summarised in the flowchart below, followed by the corresponding Minitab screenshots for the design-generation steps.

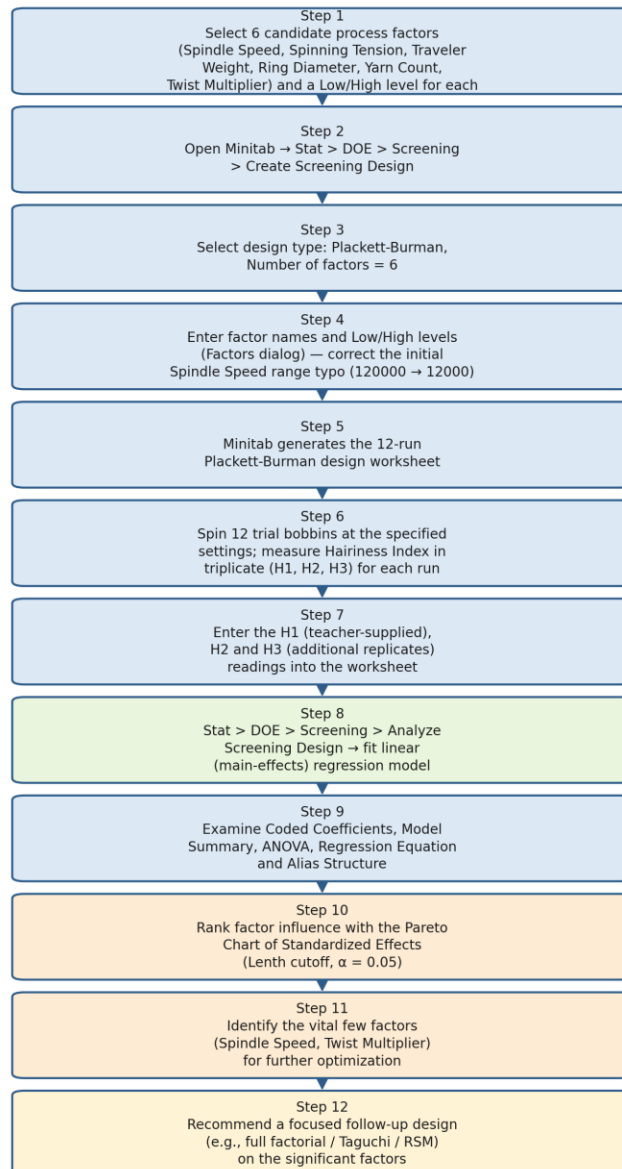


Figure 1: Step-by-step process flow for the Plackett–Burman screening study

4.1 Minitab Session — Design Generation Screenshots

Step 2–3: A new Minitab worksheet was opened and the screening design tool was invoked via Stat > DOE > Screening > Create Screening Design, selecting the Plackett-Burman design type with 6 factors.

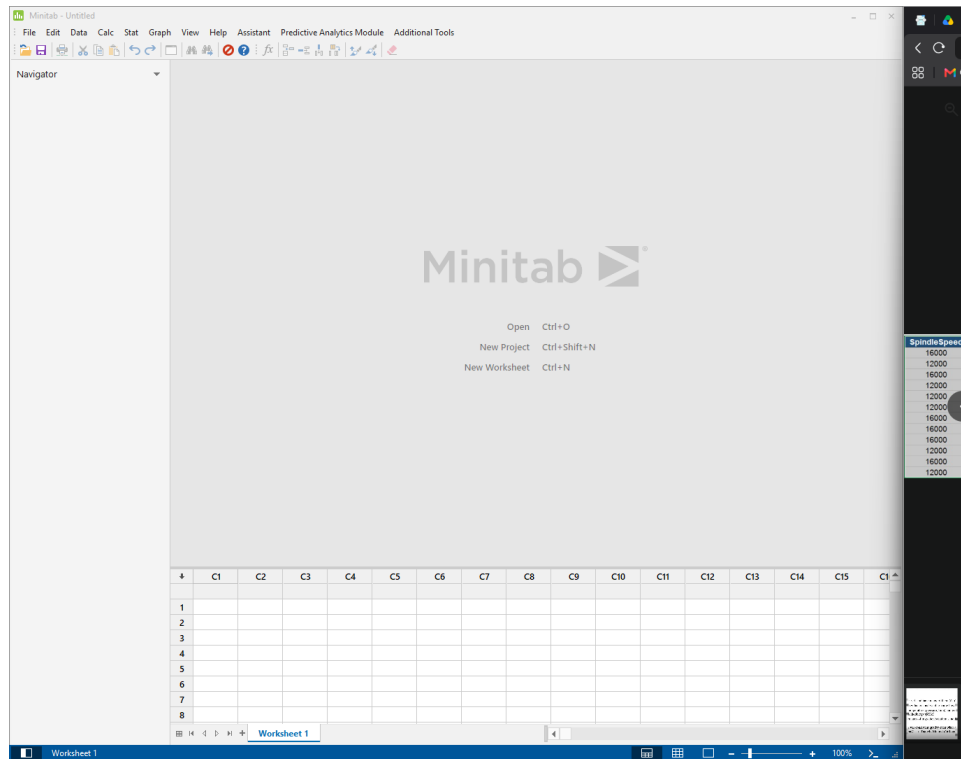


Figure 2: Minitab worksheet ready for the screening design

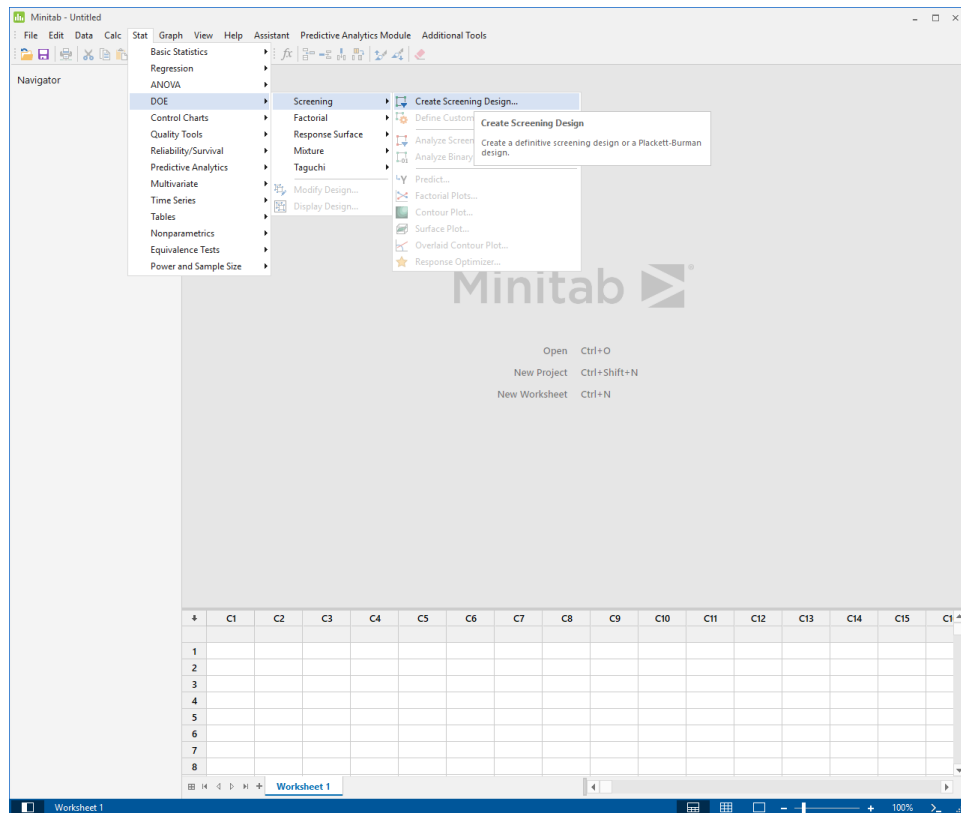


Figure 3: Stat > DOE > Screening > Create Screening Design menu path

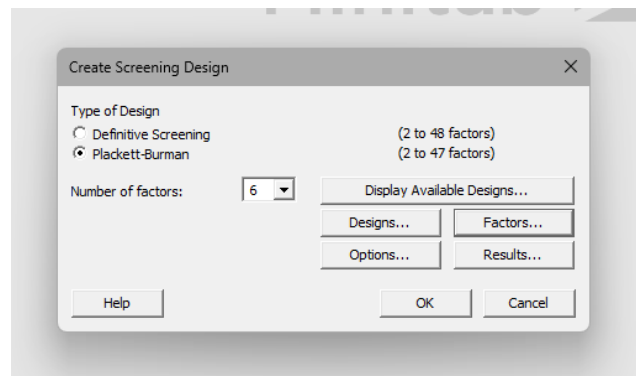


Figure 4: Create Screening Design dialog — Plackett-Burman, 6 factors

Step 4: The six factors and their level values were entered in the Factors dialog. The first attempt contained a typing error in the Spindle Speed low level (120000 instead of 12000):

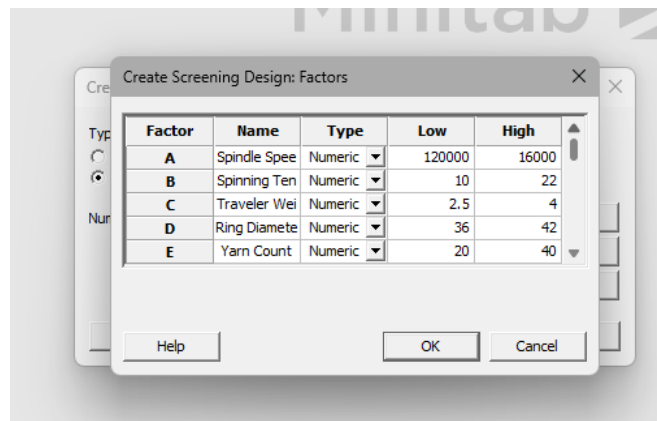


Figure 5: Factors dialog — initial entry with the Spindle Speed low-level typo

This was corrected before generating the design:

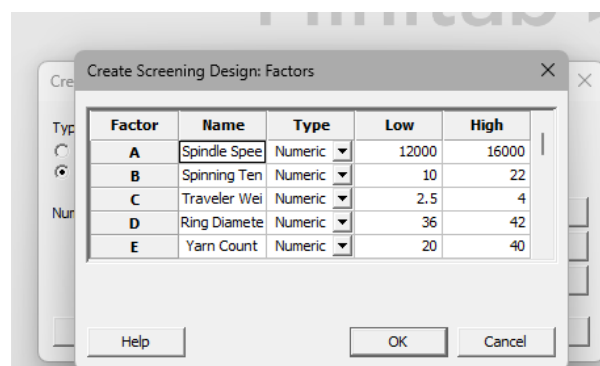
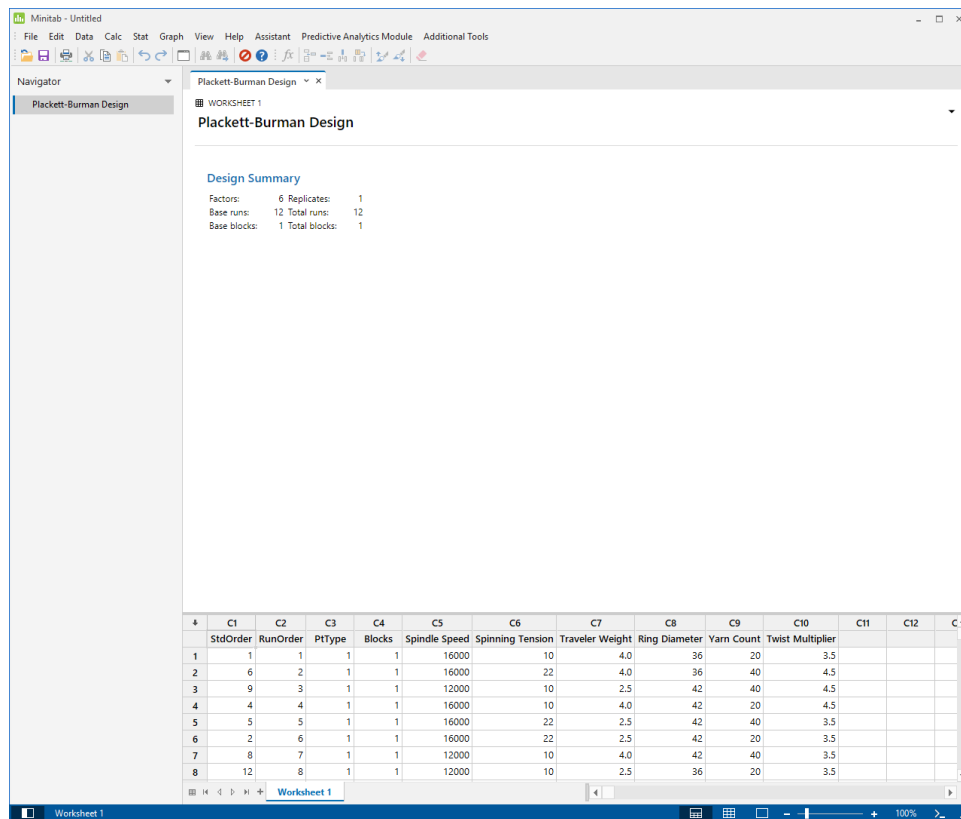


Figure 6: Factors dialog — corrected Spindle Speed range (12000–16000 rpm)

5. Experimental Design — Plackett–Burman (12-Run) Array

Step 5: Minitab generated the 12-run Plackett–Burman design worksheet shown below, giving the actual (uncoded) factor-level combination to be run in each trial.



Design Summary

Factors: 6 Replicates: 1
Base runs: 12 Total runs: 12
Base blocks: 1 Total blocks: 1

#	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
	StdOrder	RunOrder	PtType	Blocks	Spindle Speed	Spinning Tension	Traveler Weight	Ring Diameter	Yarn Count	Twist Multiplier		
1	1	1	1	1	16000	10	4.0	36	20	3.5		
2	6	2	1	1	16000	22	4.0	36	40	4.5		
3	9	3	1	1	12000	10	2.5	42	40	4.5		
4	4	4	1	1	16000	10	4.0	42	20	4.5		
5	5	5	1	1	16000	22	2.5	42	40	3.5		
6	2	6	1	1	16000	22	2.5	42	20	3.5		
7	8	7	1	1	12000	10	4.0	42	40	3.5		
8	12	8	1	1	12000	10	2.5	36	20	3.5		

Figure 7: Minitab-generated 12-run Plackett–Burman design worksheet

Run	A: Spindle Speed (rpm)	B: Spinning Tension (cN)	C: Traveler Weight (mg)	D: Ring Diameter (mm)	E: Yarn Count (Ne)	F: Twist Multiplier
1	16000	10	4	36	20	3.5
2	16000	22	4	36	40	4.5
3	12000	10	2.5	42	40	4.5
4	16000	10	4	42	20	4.5
5	16000	22	2.5	42	40	3.5
6	16000	22	2.5	42	20	3.5
7	12000	10	4	42	40	3.5
8	12000	10	2.5	36	20	3.5
9	16000	22	4	42	20	4.5
10	12000	22	4	36	40	3.5
11	12000	22	2.5	36	20	4.5
12	12000	10	2.5	36	40	4.5

6. Experimental Data / Results

Step 6–7: Three replicate Hairiness Index readings (H1, H2, H3) were recorded for each of the 12 trials. H1 was the reference value supplied by the instructor; H2 and H3 were additional replicates generated by the student, as illustrated below.

SpindleSpeed	SpinningTension	TravelerWeight	RingDiameter	YarnCount	TwistMultiplier	HairinessIndex_H
16000	22	2.5	42	40	4.5	4.13
12000	22	4	36	40	4.5	3.79
16000	10	4	42	20	4.5	4.54
12000	22	2.5	42	40	3.5	4.09
12000	10	4	36	40	4.5	3.59
12000	10	2.5	42	20	4.5	3.29
16000	10	2.5	36	40	3.5	4.49
16000	22	2.5	36	20	4.5	4.15
16000	22	4	36	20	3.5	5.17
12000	22	4	42	20	3.5	4.63
16000	10	4	42	40	3.5	4.97
12000	10	2.5	36	20	3.5	3.77

Figure 8: Instructor-supplied reference dataset (Spindle Speed ... Twist Multiplier, Hairiness Index H1)

C11	C12	C13
Hariness Index H1	Hariness Index H2	Hariness Index H3
4.13	3.550	3.21
3.79	4.210	4.90
4.54	4.780	5.26
4.09	5.210	3.64
3.59	5.170	4.15
3.29	3.250	3.17
4.49	4.170	5.17
4.15	5.664	4.78
5.17	3.890	5.64
4.63	2.170	4.17
4.97	4.640	5.28
3.77	2.640	5.64

Figure 9: Hairiness Index replicate columns H1, H2, H3 entered in Minitab

Run	H1	H2	H3	Mean Hairiness Index
1	4.13	3.55	3.21	3.630
2	3.79	4.21	4.9	4.300
3	4.54	4.78	5.26	4.860
4	4.09	5.21	3.64	4.313
5	3.59	5.17	4.15	4.303
6	3.29	3.25	3.17	3.237
7	4.49	4.17	5.17	4.610
8	4.15	5.664	4.78	4.865
9	5.17	3.89	5.64	4.900
10	4.63	2.17	4.17	3.657
11	4.97	4.64	5.28	4.963
12	3.77	2.64	5.64	4.017

7. Screening Analysis (Factorial Regression)

A linear (main-effects-only) regression model was fitted in Minitab via Stat > DOE > Screening > Analyze Screening Design, as shown below, to quantify each factor's effect on hairiness. Separate models were fitted for each replicate (H1, H2, H3); the full detailed output captured for this report corresponds to the H3 replicate and is presented below as the representative analysis.

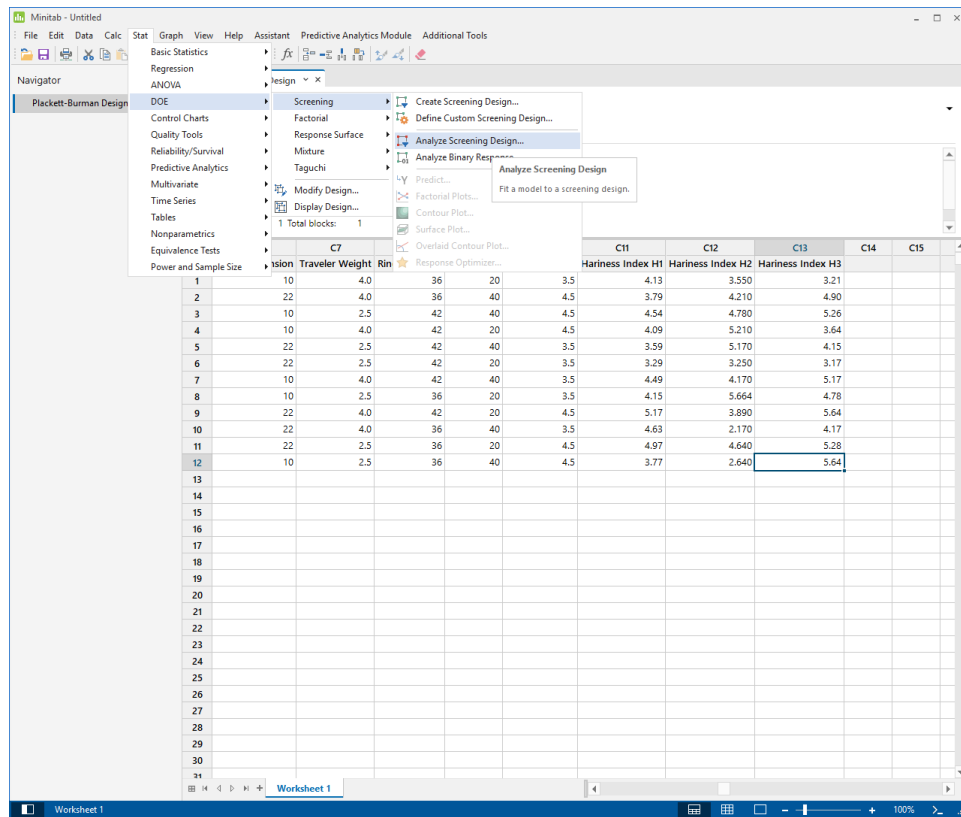


Figure 10: Stat > DOE > Screening > Analyze Screening Design menu, with the complete 12-run data set (H1–H3) visible

7.1 Coded Coefficients and Model Summary

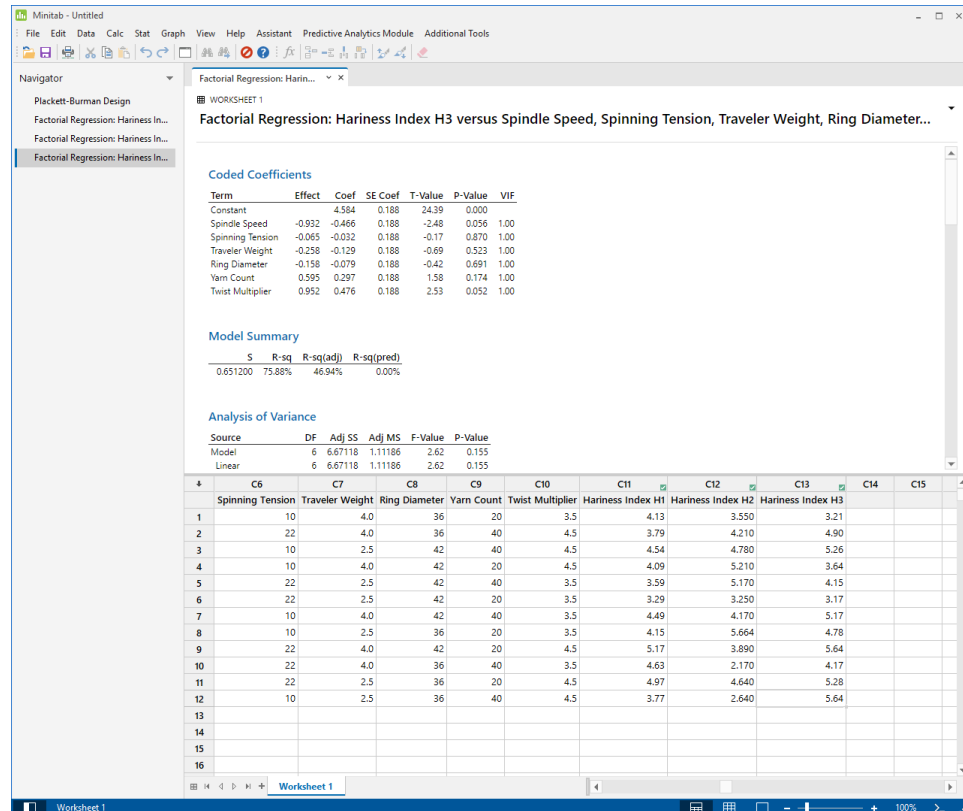


Figure 11: Factorial Regression output — Hairiness Index H3 versus the six process factors

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant	—	4.584	0.188	24.39	0.000	—
A: Spindle Speed	−0.932	−0.466	0.188	−2.48	0.056	1.00
B: Spinning Tension	−0.065	−0.032	0.188	−0.17	0.870	1.00
C: Traveler Weight	−0.258	−0.129	0.188	−0.69	0.523	1.00
D: Ring Diameter	−0.158	−0.079	0.188	−0.42	0.691	1.00
E: Yarn Count	0.595	0.297	0.188	1.58	0.174	1.00
F: Twist Multiplier	0.952	0.476	0.188	2.53	0.052	1.00

S	R-sq	R-sq (adj)	R-sq (pred)
0.651200	75.88%	46.94%	0.00%

The VIF of 1.00 for every term confirms the design's orthogonality — each factor's effect is estimated completely independently of the others. R-sq = 75.88% looks encouraging, but R-sq(adj) drops sharply to 46.94% and R-sq(pred) collapses to 0.00%, indicating that the saturated 6-term model is over-fitting this small (12-run) data set and has essentially no predictive power on new data. This is expected and acceptable at the screening stage — the purpose here is only to rank and flag candidate factors, not to build a final predictive model.

7.2 Analysis of Variance (ANOVA)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	6	6.67118	1.11186	2.62	0.155
Linear	6	6.67118	1.11186	2.62	0.155
Spindle Speed	1	2.60401	2.60401	6.14	0.056
Spinning Tension	1	0.01268	0.01268	0.03	0.870
Traveler Weight	1	0.20021	0.20021	0.47	0.523
Ring Diameter	1	0.07521	0.07521	0.18	0.691
Yarn Count	1	1.06207	1.06207	2.50	0.174
Twist Multiplier	1	2.71701	2.71701	6.41	0.052
Error	5	2.12031	0.42406		
Total	11	8.79149			

Figure 12: ANOVA for the Hairiness Index H3 linear model

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	6	6.67118	1.11186	2.62	0.155
Linear	6	6.67118	1.11186	2.62	0.155
Spindle Speed	1	2.60401	2.60401	6.14	0.056
Spinning Tension	1	0.01268	0.01268	0.03	0.870
Traveler Weight	1	0.20021	0.20021	0.47	0.523
Ring Diameter	1	0.07521	0.07521	0.18	0.691
Yarn Count	1	1.06207	1.06207	2.50	0.174
Twist Multiplier	1	2.71701	2.71701	6.41	0.052
Error	5	2.12031	0.42406		
Total	11	8.79149			

The overall model is not statistically significant at the conventional $\alpha = 0.05$ level (Model $P = 0.155$), which is common for saturated screening models with only 5 error degrees of freedom. However, individually, Spindle Speed ($P = 0.056$) and Twist Multiplier ($P = 0.052$) are both very close to the 0.05 threshold and carry by far the largest sums of squares (2.604 and 2.717 respectively, together accounting for over 79% of the model's explained variation), clearly separating them from the remaining four factors.

7.3 Regression Equation (Uncoded Units)

Regression Equation in Uncoded Units

$$\text{Hairiness Index H3} = 4.82 - 0.000233 \text{ Spindle Speed} - 0.0054 \text{ Spinning Tension} \\ - 0.172 \text{ Traveler Weight} - 0.0264 \text{ Ring Diameter} + 0.0297 \text{ Yarn Count} \\ + 0.952 \text{ Twist Multiplier}$$

Figure 13: Regression equation in uncoded units

$$\text{Hairiness Index H3} = 4.82 - 0.000233 \times \text{Spindle Speed} - 0.0054 \times \text{Spinning Tension} - 0.172 \times \text{Traveler Weight} - \\ 0.0264 \times \text{Ring Diameter} + 0.0297 \times \text{Yarn Count} + 0.952 \times \text{Twist Multiplier}$$

The signs of the equation indicate that hairiness decreases as Spindle Speed, Spinning Tension, Traveler Weight and Ring Diameter increase, while it increases with Yarn Count and, most strongly, with Twist Multiplier.

7.4 Alias Structure

Because the Plackett-Burman design is Resolution III, every main effect is aliased with a specific combination of two-factor and higher-order interactions, as shown in the alias structure below. This means a factor's apparent significance could, in principle, be partly due to an aliased interaction rather than a pure main effect — a limitation inherent to all 12-run screening designs and the reason a confirmatory follow-up design is recommended in Section 9.

Alias Structure

Factor Name	
A	Spindle Speed
B	Spinning Tension
C	Traveler Weight
D	Ring Diameter
E	Yarn Count
F	Twist Multiplier

Aliases	
I - 0.33 ABC + 0.33 ABD + 0.33 ABE + 0.33 ABF - 0.33 ACD - 0.33 ACE + 0.33 ACF - 0.33 ADE	
- 0.33 ADF + 0.33 AEF - 0.33 BCD + 0.33 BCE + 0.33 BCF - 0.33 BDE - 0.33 BDF - 0.33 BEF	
- 0.33 CDE + 0.33 CDF - 0.33 CEF - 0.33 DEF - 0.33 ABCD + 0.33 ABCE + 0.33 ABCF + 0.33 ABDE	
- 0.33 ABDF + 0.33 ABEF + 0.33 ACDE + 0.33 ACDF + 0.33 ACEF - 0.33 ADEF - 0.33 BCDE	
+ 0.33 BCDF + 0.33 BCEF + 0.33 BDEF - 0.33 CDEF - 0.67 ABCDF	
A - 0.33 BC + 0.33 BD + 0.33 BE - 0.33 BF - 0.33 CD - 0.33 CE + 0.33 CF - 0.33 DE - 0.33 DF	
+ 0.33 EF - 0.33 BCD + 0.33 BCE + 0.33 BCF + 0.33 BDE - 0.33 BDF + 0.33 BEF + 0.33 CDE	
+ 0.33 CDF + 0.33 CEF - 0.33 DEF - 0.33 ABCD + 0.33 ABCE + 0.33 ABCF - 0.33 ABDE - 0.33 ABDF	
- 0.33 ABEF - 0.33 ACDE + 0.33 ACDF - 0.33 ACEF - 0.33 ADEF - 0.67 BCDF - 0.33 ABCDE	
+ 0.33 ABCDF + 0.33 ABCEF + 0.33 ABDEF - 0.33 ACDEF	
B - 0.33 AC + 0.33 AD + 0.33 AE - 0.33 AF - 0.33 CD + 0.33 CE + 0.33 CF - 0.33 DE - 0.33 DF	
- 0.33 EF - 0.33 ACD + 0.33 ACE + 0.33 ACF + 0.33 ADE - 0.33 ADF + 0.33 AEF - 0.33 CDE	
+ 0.33 CDF + 0.33 CEF + 0.33 DEF - 0.33 ABCD - 0.33 ABCE + 0.33 ABCF - 0.33 ABDE - 0.33 ABDF	
+ 0.33 ABEF - 0.67 ACDF - 0.33 BCDE + 0.33 BCDF - 0.33 BCEF - 0.33 BDEF + 0.33 ABCDE	
+ 0.33 ABCDF + 0.33 ABCEF - 0.33 ABDEF - 0.33 BCDEF	
C - 0.33 AB - 0.33 AD - 0.33 AE + 0.33 AF - 0.33 BD + 0.33 BE + 0.33 BF - 0.33 DE + 0.33 DF	
- 0.33 EF - 0.33 ABD + 0.33 ABE + 0.33 ABF + 0.33 ADE + 0.33 ADF + 0.33 AEF - 0.33 BDE	
+ 0.33 BDF + 0.33 BEF - 0.33 DEF + 0.33 ABCD + 0.33 ABCE - 0.33 ABCF - 0.67 ABDF - 0.33 ACDE	
- 0.33 ACDF + 0.33 ACEF - 0.33 BCDE - 0.33 BCDF - 0.33 BCEF - 0.33 CDEF + 0.33 ABCDE	
- 0.33 ABCDF + 0.33 ABCEF - 0.33 ACDEF + 0.33 BCDEF	
D - 0.33 AB - 0.33 AC - 0.33 AE - 0.33 AF - 0.33 BC - 0.33 BE - 0.33 BF - 0.33 CE + 0.33 CF	
- 0.33 EF - 0.33 ABC + 0.33 ABE - 0.33 ABF + 0.33 ACE + 0.33 ACF - 0.33 AEF - 0.33 BCE	
+ 0.33 BCF + 0.33 BEF - 0.33 CEF - 0.33 ABCD - 0.67 ABCF + 0.33 ABDE - 0.33 ABDF - 0.33 ACDE	
+ 0.33 ACDF + 0.33 ADEF + 0.33 BCDE + 0.33 BCDF - 0.33 BDEF - 0.33 CDEF + 0.33 ABCDE	
+ 0.33 ABCDF + 0.33 ABDEF + 0.33 ACDEF + 0.33 BCDEF	
E - 0.33 AB - 0.33 AC - 0.33 AD + 0.33 AE + 0.33 BC - 0.33 BD - 0.33 BE + 0.33 CD - 0.33 CF	
- 0.33 DF + 0.33 ABC + 0.33 ABD + 0.33 ABF + 0.33 ACD + 0.33 ACF - 0.33 ADF - 0.33 BCD	
+ 0.33 BCF + 0.33 BDF - 0.33 CDF - 0.33 ABCE + 0.33 ABDE - 0.33 ABEF - 0.33 ACDE + 0.33 ACEF	
- 0.33 ADEF - 0.33 BCDE + 0.33 BCEF - 0.33 BDEF + 0.33 CDEF - 0.33 ABCDE + 0.33 ABCEF	
- 0.33 ABDEF + 0.33 ACDEF + 0.33 BCDEF - 0.67 ABCDEF	
F - 0.33 AB + 0.33 AC - 0.33 AD + 0.33 AE + 0.33 BC - 0.33 BD - 0.33 BE + 0.33 CD - 0.33 CE	
- 0.33 DE + 0.33 ABC - 0.33 ABD + 0.33 ABE + 0.33 ACD + 0.33 ACE - 0.33 ADE + 0.33 BCD	
+ 0.33 BCE + 0.33 BDE - 0.33 CDE - 0.67 ABCD - 0.33 ABCF + 0.33 ABDF + 0.33 ABEF - 0.33 ACDF	
- 0.33 ACEF - 0.33 ADEF - 0.33 BCDF + 0.33 BCEF - 0.33 BDEF - 0.33 CDEF - 0.33 ABCDF	
+ 0.33 ABCEF + 0.33 ABDEF + 0.33 ACDEF - 0.33 BCDEF	

Figure 14: Alias structure for the 12-run, 6-factor Plackett–Burman design

7.5 Pareto Chart of Standardized Effects

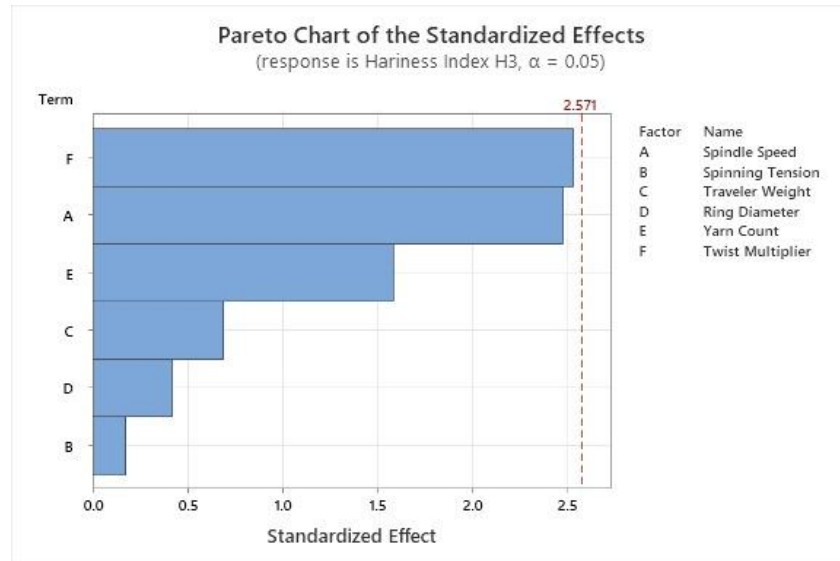


Figure 15: Pareto chart of standardized effects (response: Hairiness Index H3, $\alpha = 0.05$)

The Pareto chart ranks the six factors by the absolute standardized effect ($|t|$): Twist Multiplier (F) and Spindle Speed (A) are the two longest bars, both extending to the Lenth reference line at 2.571, followed by Yarn Count (E), Traveler Weight (C), Ring Diameter (D) and Spinning Tension (B), the last of which is negligible.

8. Identification of Significant Factors

Combining the P-values, effect magnitudes, ANOVA sums of squares and the Pareto chart, the six factors can be grouped as follows:

Factor	Effect	P-Value	Verdict
F: Twist Multiplier	0.952	0.052	Marginally significant — strongest effect
A: Spindle Speed	0.932	0.056	Marginally significant — close second
E: Yarn Count	0.595	0.174	Not significant, but notable
C: Traveler Weight	0.258	0.523	Not significant
D: Ring Diameter	0.158	0.691	Not significant
B: Spinning Tension	0.065	0.870	Negligible

Twist Multiplier and Spindle Speed emerge as the ‘vital few’ factors governing yarn hairiness in this study, together explaining roughly 79% of the model's adjusted sum of squares, while Spinning Tension, Ring Diameter and Traveler Weight behave as the ‘trivial many’ and can be fixed at whichever level is most convenient for production without materially affecting hairiness. Yarn Count shows a moderate, non-significant trend that may deserve inclusion in a follow-up study.

9. Recommended Follow-Up Study

Since this is a Resolution III screening design, the main effects of Twist Multiplier and Spindle Speed are aliased with two-factor interactions (Section 7.4), and neither factor is significant at a strict $\alpha = 0.05$ with only 5 error degrees of freedom. Before finalizing process settings, it is recommended to:

- Run a full 2^2 (or 2^3 , including Yarn Count) factorial design, or a Response Surface / Taguchi optimization design, on Spindle Speed and Twist Multiplier (and optionally Yarn Count) to obtain de-aliased, precise estimates of their main effects and interaction.
- Fix Spinning Tension, Traveler Weight and Ring Diameter at convenient mid-range production settings, since they showed negligible effect on hairiness in this screening study.
- Increase the number of replicates per run (e.g., 3–5) in the follow-up study to shrink the error variance and improve the power to detect the Spindle Speed and Twist Multiplier effects conclusively.

10. Conclusion

A 12-run Plackett–Burman screening design was used to evaluate the effect of six ring-spinning process parameters on yarn Hairiness Index. Based on the Factorial Regression, ANOVA and Pareto chart analysis, the following conclusions can be drawn:

1. Most Influential Factors: Twist Multiplier (Effect = 0.952, $P = 0.052$) and Spindle Speed (Effect = -0.932 , $P = 0.056$) are the two dominant factors affecting hairiness, together accounting for the majority of the explained variation (ANOVA Adj SS of 2.717 and 2.604 respectively, versus a total model Adj SS of 6.671).
2. Direction of Effects: Hairiness increases with increasing Twist Multiplier and decreases with increasing Spindle Speed, based on the signs of their coefficients (+0.476 and -0.466 respectively) in the fitted model.
3. Minor Factors: Yarn Count showed a moderate but non-significant trend ($P = 0.174$), while Traveler Weight, Ring Diameter and especially Spinning Tension ($P = 0.870$) had negligible effect on hairiness within the ranges tested.
4. Model Limitations: The saturated main-effects model achieved $R\text{-sq} = 75.88\%$ but $R\text{-sq}(\text{pred}) = 0.00\%$, and the design's Resolution III alias structure means main effects are confounded with two-factor interactions — both expected characteristics of a 12-run screening design that should not be over-interpreted as a final predictive model.
5. Recommendation: A focused, higher-resolution follow-up experiment (full factorial, Response Surface, or Taguchi design) on Spindle Speed and Twist Multiplier is recommended to obtain de-aliased estimates and determine the optimal settings for minimum yarn hairiness.

Overall, the Plackett–Burman screening design successfully reduced the experimental effort from 64 runs (full 2^6 factorial) to just 12 runs while still clearly separating the two most influential process parameters from the four unimportant ones, demonstrating the practical value of screening designs as the first stage of a structured process-optimization programme.